

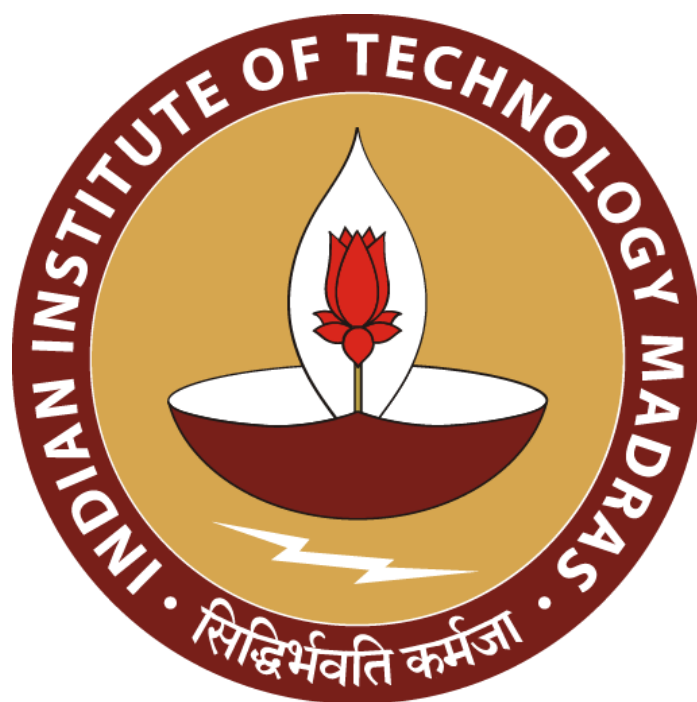
# Enhancing Real Estate Business Through Data Driven Analysis

BDM CAPSTONE PROJECT MID-TERM SUBMISSION

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(BS) DEGREE IN DATA SCIENCE AND APPLICATIONS

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# Executive Summary

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**Proprietor:** Mr. Rakesh Narang

**Organization:** Narang Associates

**Site:** S-17B, Tent Market, Indira Nagar, Aadarsh Nagar, Delhi-110033

**Employees working:** 4

**Operating Hours:** 10:00 AM - 7:00 PM, Monday to Saturday

**Years in Operation:** 10 Years

Narang Associates is a small real estate business with a location in Indira Nagar and Aadarsh Nagar, Delhi. It is facing huge issues in terms of market reach and increasing numbers for profit margins. The company is working in a highly competitive environment and mostly pursuing traditional ways of business that will not be beneficial in the growing prospects for the company. Strategic data analysis will go miles in providing actionable insights for this small real estate company headed by Narang Associates, towards which this mid-term business report seeks to target ways out.

The business model of Narang Associates is commission-driven, with property sales and rentals being handled primarily. It is in the area of data-driven decision-making that forms one of the major impediments to growth. This report will seek to analyse the primary data collected concerning market dynamics, customer demographics, and deal types of the business using a different set of statistical measures and visualization tools. Descriptive and inferential statistics, correlation analysis, and market segmentation will be applied in the analysis to arrive at an appropriate interpretation of the business landscape.

The expected results are the determination of new markets for expansion and changing the marketing efforts for the attainment of greater returns through high-commission deals, which brings better profitability and growth. Narang Associates bank on the insights from data to extricate themselves from the current challenges, which assure continuous growth and effective realization of better market positioning.

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## Proof of Originality

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To establish the authenticity of the data, the supporting evidence as listed below:

- **Principal Data Source:** Mr. Rakesh Narang and Ayush Narang were interviewed face-to-face in order to gather data. These interviews gave insightful information about the workings of the company.
- **Video Interaction:** A video interview with Mr. Ayush Narang covering the operations and business difficulties. You can use the following G-Drive URL to access the video interaction: [Link 1](#) [Link 2](#)
- **Communication from the Association:** Included is a signed letter from Mr. Ayush Narang attesting to their cooperation and supply of data for this project. You can access the letter by clicking the G-Drive link [Letter from Narang Associates](#)
- **Organizational Images:** The organization's images. Access to these photos is also possible via the G-Drive link [Images of the Organization](#), which offers visual context for the business environment

- **Photograph of the Organization's Business Card:** The accompanying Google Drive link [Business Card Image](#) allows you to view the business card as well
- **Emails from the Owner:** Emails exchanged with the owner of Narang Associates, supported by official student identification, serve as confirmation of the authenticity of the data collected. [Email Correspondence](#)

## Metadata and Descriptive Statistics

### METADATA:

The owner generously granted me access to varied data sources as listed:

#### Data Description:

The owner has shared a complied data of 4 years in excel sheet containing details about property they deal with and clients demographics which contains property type, deal type, property location, reason for buying, amount of deal, deal source, commission earned, expenses incurred, client's gender, client's age, numbers of dealer involved in a particular deal.

- **Property Type:** what kind of property they are dealing with a client, whether it is plot, flat, shop.
- **Deal Type:** whether the deal is for rent or sale to the client.
- **Property Location:** where the property is located. The business owner deals at Aadrash Nagar, Azadpur, Indira Nagar, Kewal Park, Majlis Park, Model town, Mukundpur.
- **Reason for Buying:** why the client is renting or purchasing the property whether it's for investment or residential.
- **Amount of Deal:** Amount of property.
- **Deal Source:** deal source is referral or walk-in.
- **Commission Earned:** how much commission does he earns through deal
- **Client Gender:** client's gender (male or female)
- **Client Age:** what is the range of age of client.
- **Number of Dealers Involved:** how many deals he made yearly, monthly
- **Expense Incurred:** expenses business incurred while making deals.

### Clients Data Metadata Keys:

| Key             | Description                            |
|-----------------|----------------------------------------|
| Client's Gender | Which gender is most frequent buyer    |
| Client's Age    | Which age group is most frequent buyer |

### Deals Data Metadata Keys:

| Keys                        | Description                                                                                                                  |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Amount of Deal              | INR value of property                                                                                                        |
| Commission Earned           | The INR value of commission earned by the business owner.                                                                    |
| Deal Source                 | From which source the clients are coming to him                                                                              |
| Number Of Dealers Involved. | how many dealers are involved in a deal                                                                                      |
| Expenses Incurred           | How much money does they spent on a client, rent of office, marketing expenses, salary of staff, electricity and water bill. |

## DESCRIPTIVE STATISTICS:

After data processing and cleansing, the following is a concise overview of the dataset using descriptive statistics. I have attempted to condense the information by highlighting the descriptive stats below relevant for both deals and client's data.

| Descriptive statistic measure | Descriptive statistic definition                            |
|-------------------------------|-------------------------------------------------------------|
| Sum                           | The total value.                                            |
| Mean                          | The average value.                                          |
| Median                        | The middle value when arranged in ascending order.          |
| Standard Deviation            | The measure of spread of values from mean in single sample. |
| Minimum                       | The smallest value.                                         |
| Maximum                       | The largest value.                                          |

### Descriptive Statistics Summary of Deals Data:

| Descriptive statistic measure | Amount of deal  | Commission earned from buyer | Commission earned from seller | Expenses incurred |
|-------------------------------|-----------------|------------------------------|-------------------------------|-------------------|
| Sum                           | Rs.25,01,53,000 | Rs.45,81,500                 | Rs. 42,81,500                 | Rs. 24,3345.368   |
| Mean                          | Rs.11,68,940    | Rs. 21,409                   | Rs. 20,007                    | Rs. 1,131.83      |
| Median                        | Rs.20,000       | Rs.10,000                    | Rs. 10,000                    | Rs. 1,000         |
| Standard Deviation            | Rs.47,13,007    | Rs. 43,116.8                 | Rs. 38,986.17                 | ₹ 645.37          |
| Minimum                       | Rs.12,000       | Rs. 0                        | Rs. 0                         | Rs. 600           |
| Maximum                       | Rs. 3,00,00,000 | Rs. 3,00,000                 | Rs. 3,00,000                  | Rs. 9000          |

The property deals have been organized into 7 locations for the purpose of analysis. These locations are as follows:

1. Azadpur = Renting and sales of plots, flats and shops. And commission earned from there
2. Indira Nagar = Renting and sales of plots, flats and shops. And commission earned from there.
3. Kewal Park = Renting and sales of plots, flats and shops. And commission earned from there
4. Majlis Park = Renting and sales of plots, flats and shops. And commission earned from there
5. Model Town = Renting and sales of plots, flats and shops. And commission earned from there
6. Mukundpur = Renting and sales of plots, flats and shops. And commission earned from there
7. Aadarsh Nagar = Renting and sales of plots, flats and shops. And commission earned from there

The focus of the analysis will be directed towards the top locations where the business gets its most of deals. The location wise descriptive analysis can be accessed through the following link: [descriptive stats location.xlsx](#)

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# Analysis processes and methods

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The analysis of data underwent some crucial steps that were well-planned and executed to ensure that comprehensive insights about the real estate business at hand were derived. Acquiring raw data was the first step of this process, which was painstakingly gathered from different sources directly by interviewing the stakeholders at Narang Associates. Since this was the initial step, it was quite important in making sure that the data was relevant, accurate, and comprehensive enough to form a good base for analysis.

## Data Collection and Organization:

Firstly, deals per location were to be compiled, and this was the first step of analysis. In a table, data for each location was organized, further divided into rent and sale and these in turn were subdivided into property types—plot, flat, shops. It would give a very granular view of the business' performance across different property types and locations.

**Justification:** Organization of data by location and property type allowed focused insight into which areas and property types were most profitable. This is chosen over more generalized approaches for the detailed breakup to be able to develop targeted strategy development in specific areas.

## Calculation of financial metrics:

Some of the key financial metrics calculated for any location are the commission earned from the buyer and seller, expenses incurred, total deal amount, total commission, and net profit. These give an all-rounded view of the financial performance for each deal and insight into its profitability and cost efficiency.

**Justification:** These metrics needed to be computed to realize the financial health of the business. Other methods, just looking at the total revenues or profit margins, would not give so much insight into detailed performance for the individual deals and locations.

## Graphical Data Visualization:

Data visualization was an integral part of analysis. Different chart types, like bar charts, pie charts, and stacked bar charts, were used for the count of property types, location-wise deals, commission earned, amount of deal, expenses incurred, and rent vs. sale distribution.

- **Bar Charts:** These were majorly used to represent the count of various property types and their distribution across locations.
- **Pie charts** show the proportion of rent versus sale deals, providing in an instant, an integral visual summary of deal types.
- **Stacked Bar Charts:** They provided a granular view of monetary metrics, such as commission earned against deal amount, and expenses by location.

**Justification:** These visualization methodologies were used because they really brought out the clear representation of complex data in a format that the human mind could easily interpret. Line graphs or scatter plots were not used since they are less effective for categorical data and financial breakdowns.

## Correlation Analysis:

- The deal amount as a factor of correlation with the total commission earned was considered. This analysis helped in understanding the strength and direction of the relationship between these two critical financial metrics.

**Justification:** Correlation analysis was chosen to quantify the relationship between deal size and commission. Regression analysis was considered but not used as this was an exploratory analysis in nature and the focus was on finding correlations, not predicting an outcome.

### **Measure of Central Tendency:**

- All numeric data is measured for central tendency by mode, median, and mean. This summarized the distribution of the data and emphasized average deal amount, commission rate, and other critical financial metrics.

**Justification:** These were necessary steps that needed to be done in order to summarize the data and get central values around which most of the data points cluster. Other measures, like variance or standard deviations were not emphasized since the main aim was to know the central trend rather than dispersion.

### **Cluster Analysis:**

- A clustering graph was made in order to know the relationship between age and deal amount. In this way, it would be easier to find out any different patterns or groups within the data.

**Justification:** Clustering was used to find out any possible trends in their customer demographics that would affect their marketing and sales work. Other methods such as simple grouping was not applied since it would not help to discover any natural patterns within a dataset.

### **Conclusion**

- This deep analysis procedure was developed to give granular and high functionality in understanding the performance of the real estate business. Methods and tools were selected based on their capacity to squeeze meaningful information from data so that analysis will be thorough and of strategic importance. The choice of key financial metrics to focus on, together with very strong data visualization and correlation analysis, has given utmost details on the business strengths and improvement areas that can do justice to the means adopted and lay a firm base for decisions.

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## **Results and Findings**

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### **Trends and Patterns:**

• **Deal Distribution:** This would point out a huge majority in the rental transactions and specify that Indira Nagar and Majlis Park have been leading in this aspect. This then indicates that Narang Associates is hitting on a demand for rental properties in those areas.

- **Demographics:** The consumer is primarily represented by the 30 to 45 age segments, with a slight predominance of the male gender. This demographic insight is, therefore, critical in tailoring effective marketing strategies that would resound with a chief customer base and provide effective, targeted outreach.

- **Commission Patterns:** Property sales, with particular emphasis on Model Town and Indira Nagar, give much higher commissions than what rental transactions would yield. This pattern underlines that the sales transactions in such lucrative regions are more profitable.

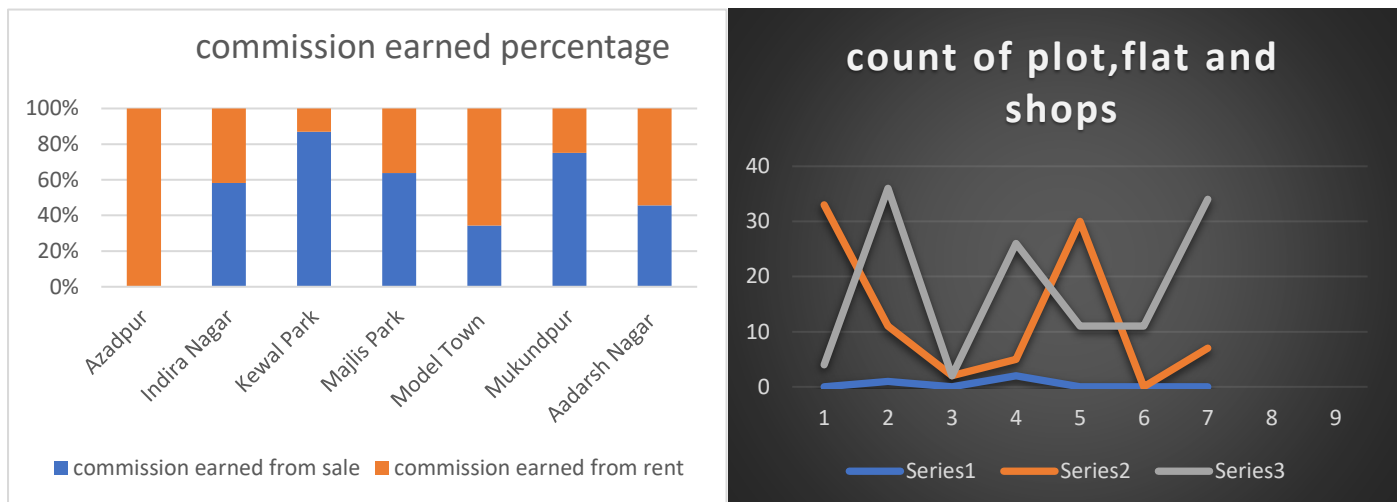
•**Increase in Expenses:** It was observed that operating expenses had risen by involving various dealers compared to one deal, which reduces net profitability. This finding pointed to the possibility of rationalizing operations and eliminating the involvement of middlemen to enhance financial efficiency.

### Graphical summaries:

•**Bars:** Used in comparison of the number of deals in various property types and locations. These offer a very clear comparative view of the business's performance in the respective segments.

•**Heat Maps:** Heat maps have successfully focused on high-value deals, outlining those areas that were very profitable and could therefore be potential zones for strategic expansion.

•**Pie Charts:** The distribution of sources of all deals is represented by pie charts, which display this visually by breaking it down to articulate the high contribution of referrals and word-of-mouth in client acquisition.



### Percentage of Commission Earned:

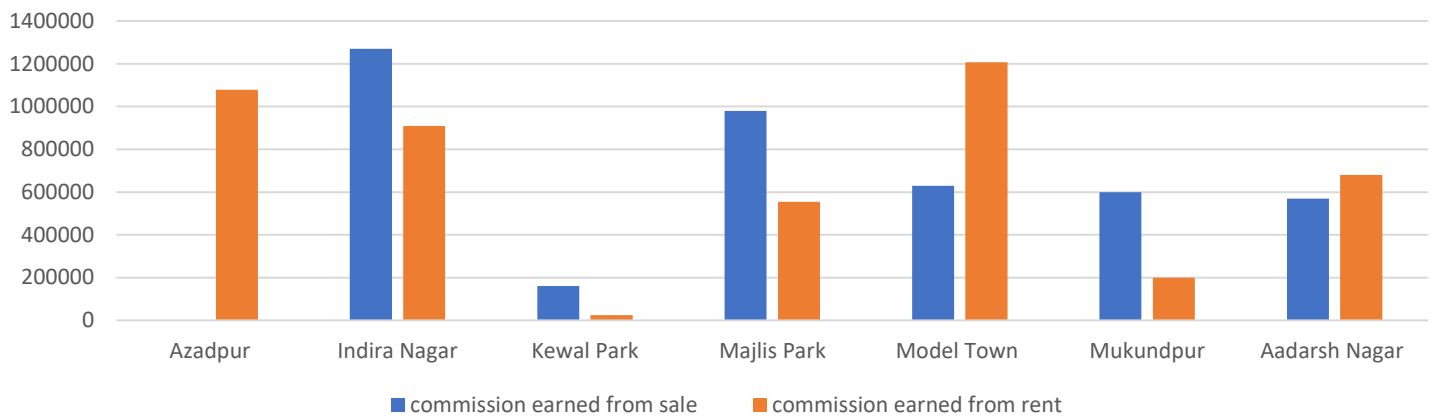
- **Insight:** The forthcoming bar chart shows the percentage of sales vs. rental commission earning at different locales.
  - Azadpur has all commissions from rentals; hence, it could be a location dominated by rental transactions.
  - Model Town and Aadarsh Nagar have an equal mix of both—commissions from sales and rentals—and hence are diversified activities in these areas.
  - Indira Nagar, Majlis Park, and Mukundpur have more percentage of commissions from rentals; hence, there is more focus or inclination towards rental properties in these locations.
  - Kewal Park has an almost equal distribution which depicts more balanced performance in the sales and rental markets.

### Count of Plot, Flat and Shops:

- **Type of Property Distribution:** This line graph indicates the count of different types of properties in respective locations or time frames.
- **Fluctuating Trends:** Ups and downs in the number of plots, flats, and shops indicate variability in popularity or availability of property types.



commission distribution

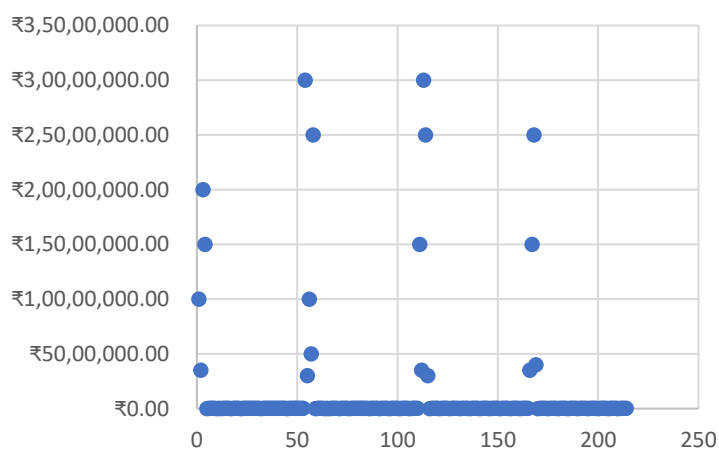


### Commission Distribution:

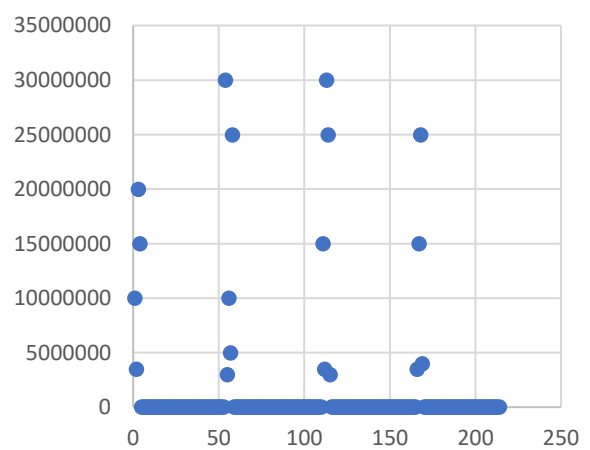
• **Insight:** Absolute values represented by the chart of the earned commission through sales and rental

- Indira Nagar shows the highest commission in sales, and the market in this region is keen for the sale of properties
- Model Town also shows high commissions on sales; hence it appears to be an area with high return or good sales transactions
- Commission rate in Azadpur is yet again very high in rentals, and the sales rate is negligible. It is thus an indication to the fact derived before; this area is focusing under rental.
- Majlis Park and Mukundpur have a moderate amount of commission earned on both sales and rentals, meaning there is relatively balanced performance in terms of sales and rentals and not a large skew toward one category.

Amount of Deal



clustering of age and amount

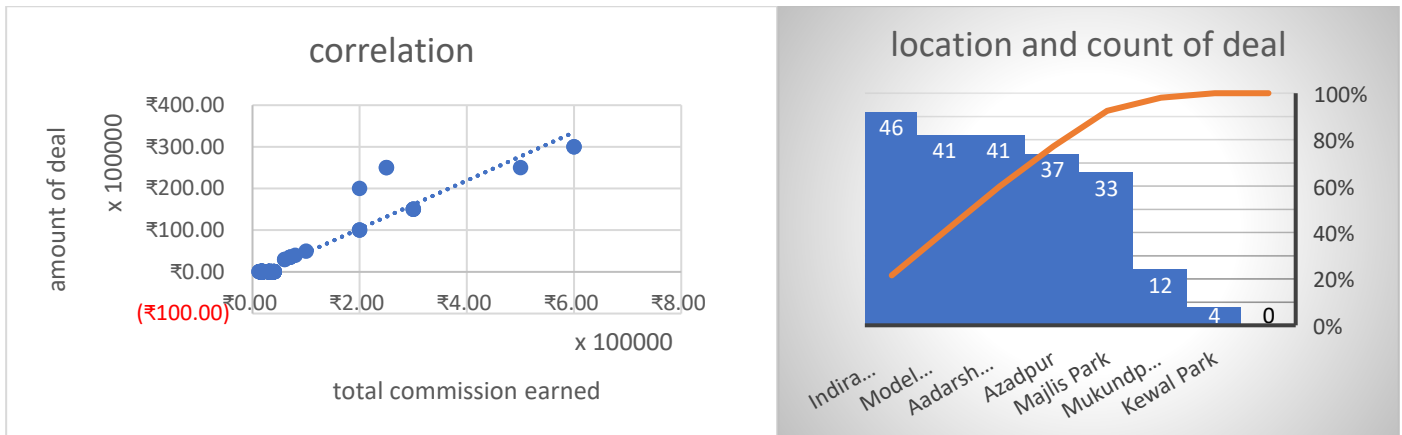


### Amount of Deal (Scatter Plot):

• **Insight:** This scatter plot indicates the deal amount distribution for all transactions.

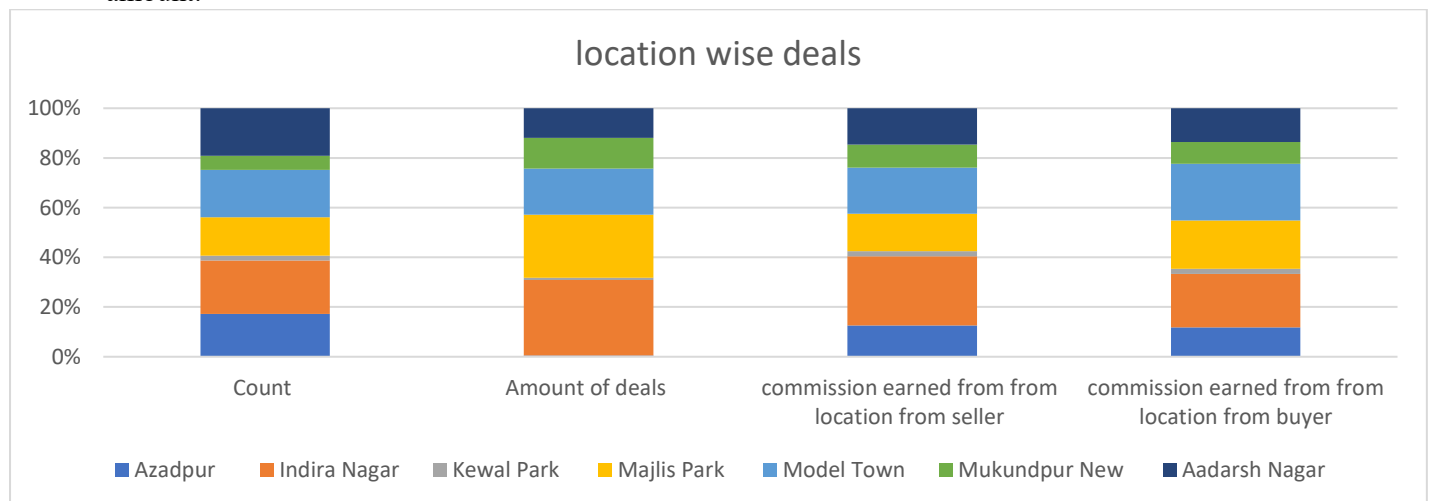
- Deals include a wide range of deal amounts, but certain high-value transactions have gone up to ₹3.5 crores or more.
- The majority of the deals are clustered below ₹50 lakhs, which goes on to prove that smaller transactions are more frequent.

- The wide spread of the points indicates deal size variance. This could be due to location, property type, and client demographic factors.
- A few high-value transactions can be considered as outliers, and these are of high value and individually contribute significantly to profitability.



### Insights from the Correlation Graph:

- **Positive Correlation:** The graph shows a positive correlation between the total commission earned and deal amount. This means that, as the deal amount goes on increasing, the earned commission also has a tendency to increase. The data points very nearly follow a linear trend, indicating a strong linear relationship between the two variables.
- **Slope of the Trend Line:** Slope of the trend line: With the increasing deal amount, the commission earned increases proportionately. The greater the slope, the more the commission would increase as a function of deal amount.



### Location Wise Deals:

- **Diverse Contribution:** The stacked bar chart indicates contribution from varied locations, color-coded, to the total number of deals, the amount of deals and the commission earned from the buyer and seller.
- **Dominance of Indira Nagar:** Indira Nagar leads in all categories, and this lead is conspicuous in the count and deal amount.
- **Location Impact on Commission:** The contribution to commission from buyers and sellers varies across locations, an indication that some locations may be more lucrative in terms of commission.

### The graphs presented above illustrate key takeaways as follows:

Sales dominate in Indira Nagar and Model Town; rent rules the roost in Azadpur. Indira Nagar and Model Town record maximum transactions, Kewal Park the least. Demand varies from place to place. Flats and shops show fluctuation. Most deals fall within specific age groups and deal amounts, with a few high-value outliers.